

- **Hindi** : Mother tongue
- **German/Deutsch**
(Goethe-Zertifikat B2)
 - Listening: B2
 - Reading: B2
 - Spoken production: B2
 - Spoken interaction: B2
 - Writing: B2
- **English (IELTS)**
 - Listening: C2
 - Reading: C2
 - Spoken production: C1
 - Spoken interaction: C2
 - Writing: C1

★ Skills

- Machine Learning & Data Science (Mathematics):
- Deep Learning: Keras & PyTorch
 - Machine Learning: scikit-learn, tensorflow
 - LLM: Transformers, RAG, Encoders, & Decoders
 - Big Data: Hadoop, Spark, Hive, Kafka
 - Data Dashboarding: MS PowerBI
 - Computer Vision: OpenCV, ITK
- DevOps & Linux SysAdmin:
- NixOS Expert
 - IaC: nix, ansible & terraform
 - Cloud Experience: AWS, GCP & Firebase
 - VCS: git, jj
 - Shells: bash, zsh, fish, pwsh
 - Virtualization & Containerization: Proxmox, KVM, Docker & Podman
 - CI/CD: GitHub Actions & Jenkins

Divit Mittal

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<https://www.kaggle.com/divitmittal> :: <https://huggingface.co/divitmittal>

🏢 Work experience

Funded Deep-Learning Research Position 15/05/2025 – 01/09/2025
 Manipal University | Jaipur, Rajasthan, India

Research on deep learning architecture combining **FocalViT** and **CrossViT** transformers for **multi-focus image fusion**. Achieved excellent fusion quality on benchmark datasets, demonstrating improved preservation of high-frequency details compared to CNN-based approaches.

<https://hf.co/spaces/divitmittal/hybridtransformer-mfif>

Open Source & Community Contributions 01/07/2024 – 01/10/2024
 Remote

Active contributor to the functional programming ecosystem, particularly within the Nix community. Participated in Nix consulting with **Tweag**, **ModusCreate**, and **Blockfrost**, providing expertise on complex NixOS configurations, package derivations, and infrastructure automation for enterprise clients.

DevOps & System Administrator 15/12/2021 – 28/02/2022
 UbiOps (Yes!Delft) | Delft, Netherlands

Administered containerized environments using **Docker** for multi-tenant ML workloads, implemented CI/CD pipelines for automated model deployment, and maintained cloud infrastructure across **AWS**.

🎓 Education & Training

Bachelor of Technology in Data Science & Engineering 10/09/2022 – Current
 Manipal University | Jaipur, Rajasthan, India

Final grade: ~8/10 | **National classification:** NCC5

Bachelor of Science in Computer Science & Engineering 01/09/2021 – 01/09/2022
 Delft University of Technology | Delft, Netherlands

Final grade: ~8/10 | **Type of credits:** ECTS | **Number of credits:** 20

Bachelor of Computing in Computer Science 01/08/2020 – 28/08/2020
 National University of Singapore | Singapore, Singapore

All India Senior School Certificate Examination 31/03/2018 – 12/07/2020
 Central Board of Secondary Education (CBSE) | New Delhi, India

Final grade: A1 : ~96% | **Address:** Delhi Public School, R.K. Puram

All India Secondary School Examination 31/03/2016 – 28/05/2018
 Central Board of Secondary Education (CBSE) | New Delhi, India

Final grade: A1 : ~97% | **Address:** Bhatnagar International School Paschim Vihar

■ Projects

OS-nixCfg: State-of-the-art Infrastructure as Code(IaC) configuration

31/03/2024 – Current

Nix, **nix-darwin**, **NixOS**, **home-manager**, etc. configurations to attain my **deterministically reproducible** OS across heterogeneous platforms, written in the **functional** "Nix" expression language. Maintains: Unified configuration for **4+** platforms (macOS via **nix-darwin**, Android via **nix-on-droid**, **NixOS**, **WSL2**) achieving complete reproducibility.

Starred by DRW employees, demonstrating **industry validation** of advanced Nix expertise.

<https://github.com/DivitMittal/OS-nixCfg>

Documentation Conversational AI @ IEEE MUJ

31/01/2024

RAG -enhanced **LLaMa-like transformer** written from scratch (aided via a pre-trained LLM) interfaced via a with a **streamlit** chat front-end. Implemented via **Qdrant** vector database on a **DigitalOcean Docker** container for assisting students on programming lanugages'/frameworks' documentations.

<https://github.com/DivitMittal/DocAssist-LLM>

Generation of "Hazard Map" for safely navigating a lunar lander 10/2023

Is implemented by utilising the sensory data from the **ISRO** 's Chandrayaan-2 TMC2 & OHRC payloads for **SIH** (Smart India Hackathon). The project entails wrangling multiple data in form of multispectral high-res. imagery & stereo-triplet panchromatic data of lunar terrain. These multiple streams of data are fused together and analysed respectively for ground truth via **SRGANs** & **DCNNs**.

🏆 Honours and Awards

BobbleAI Datathon Winner

31/08/2024

BobbleAI Keyboard, Gurgaon, Haryana

Processed raw product data, i.e. **BobbleAI Keyboard** (70M+ users) event records in the last 5 days before July 1, 2024,i.e., about 8 million rows. Extracted & transformed JSON, to preprocess, ML-filter and store as structured **Apache Parquet** on **AWS S3** via serverless **AWS Lambda** functions & **PySpark** on **AWS Glue**.

<https://github.com/DivitMittal/Datathon-BigData>

Proglint's Computer Vision 2k23 National Hackathon Runner-ups

30/09/2023

Alliance University, Anekal, Karnataka

Leveraging the high-fidelity **CARLA** simulator to create diverse driving scenarios for an **autonomous vehicle** driving system with **RGBA** & **LiDAR** sensors. Utilizing deep learning techniques, i.e. **CNNs** for **semantic segmentation** to improve the vehicle's understanding of its surroundings, & manouvre accordingly.

A **Driver Drowsiness Detection** system that uses **Siamese Neural Networks** and computer vision techniques, like haarcascade wave filters to monitor a driver's facial features in real-time and detect signs of fatigue. The system identifies potential drowsiness and enhances road safety.

<https://github.com/DivitMittal/CARLA-Autonomous-Driving>

<https://github.com/DivitMittal/Driver-Drowsiness-Detection>

International Rank 7 at the International Mathematics Olympiad

01/01/2020

The Science Olympiad Foundation, New Delhi

International Rank 49 at the International English Olympiad (IEO)

01/12/2019

The British Council Division, British High Commission, New Delhi